

Nirukt Agrawal

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Education:

University of Bath (2025 – 2030)

MEng (Hons) Aerospace Engineering, predicted 1st (1st year average: **75%**).

Thermofluids	84%	Design, Materials & Manufacturing	78%
Solid Mechanics	73%	Mechatronics	66%
Responsible Engineering	86%	Mathematics	64%

Aylesbury Grammar School – A Level Grades (2023 – 2025)

A*A*A*A – Maths, Physics, Computer Science, Further Maths.

Aylesbury Grammar School – GCSE Grades (2021 – 2023)

11 GCSEs: 7 grade 9s, 2 grade 8s and 2 grade 7s. *Additional Maths*: A (max grade).

Engineering Projects & Experiences:

Badminton Shuttlecock Launcher (extra-curricular) (Summer 2026):

- Conducted research, designed and manufactured an automatic badminton feather shuttlecock launcher to solve the problem of having no training partner. I used off-the-shelf parts and material stock (aluminium extrusion) and 3D-printed custom parts.
- Integrated multiple mechatronic systems for shuttle placement based on user position, such as camera tracking, shuttle feeding, shot power adjustment and gimbal-like aim.

Mechatronics Ultrasonic Sensor Project (curricular) (2025 – 2026):

- Developed a door-detection and person-counting system using an ultrasonic sensor with an accuracy of around 95%. Documented the full design, experimental methodology and results in a structured report, achieving 81%.

Team Bath Drones (extra-curricular) (2025 – 2026):

- Developed a CAD model based on preliminary calculations of a drone designed to a set of simplified regulations. The model was used to perform CFD to ensure airworthiness.
- Collaborated in manufacturing the drone, learning how to create a mould for carbon-fibre layup, wire-cut foam for wings, and create structural ribs in the fuselage. This further developed my knowledge of how important designing for manufacturing techniques is.

Pre-University Projects:

- Created a web-based CFD solver using python. I used a finite difference method along with a NACA aerofoil generator to explore the basics of airflow and lift. I learned about modelling assumptions, such as the CFL condition and how different models can cater for different scenarios. Overall, it helped me understand how the back-end works for simulation software, which can throw hard-to-understand errors.
- Collaborated to design an unmanned drone system that monitors plants for abnormal signs indicating disease and then neutralise them, increasing farm yields through early detection. This taught me how to apply an engineering mindset to real-world problems and won "Runner-up for Best Research" at Homerton College, Cambridge.

Awards:

- Arkwright engineering scholarship sponsored by Virgin Atlantic (2023 – 2025) which exposed me to commercial aviation industry insights via a mentor. The competitive selection process was based upon an aptitude test and interview before GCSEs.
- Won our school's Area Finals as the best business where I led as the Managing Director of Wild Cards Young Enterprise. I gained strong leadership and interpersonal skills while managing various business functions such as R&D, finance and marketing.
- Arthur Clement's Award as funding for my shuttlecock launcher project (£540).

Computing Skills:

3D CAD: Experience in Inventor Pro and Onshape. Using Inventor, I designed a working planetary gearbox as part of the first stage of a turbofan engine, along with a CAD model for a drone, with knowledge about assemblies, drawings, exploded views, Inventor Studio etc.

MATLAB: Used to code and communicate with an Arduino for data collection and response. Also used to create graphs for reports and presentations since it is more powerful than Excel.

MS Office: Intermediate to advanced level in Word and PowerPoint, used for presentations and reports. Intermediate level in Excel, used for data collection and calculations.

Work Experiences:

Virgin Atlantic UK Operations, Crawley HQ and Heathrow Hangar (2024):

- Learnt how aeroplanes are maintained through continuous repair and inspections, along with service bulletins to improve reliability. Using such documents was able to carry out maintenance procedures on an A330 and B787, learning the importance of engineering a product for its whole lifecycle (manufacture, use, maintenance, disposal).
- Understood how robust, documented procedures reduced errors during maintenance and in flight, such as the checklist of steps to be followed before removing an engine.

Freeform technologies, Buckingham (2024):

- Increased my understanding in CNC-based composite manufacturing of materials like carbon-fibre and permaglass. I set-up a 6-axis CNC machine and used it to produce a carbon-fibre part from a billet, intended for a hypercar. Adjusted speed to prevent chatter and other surface defects while reducing tool wear.
- Learnt about considerations in high-tolerance manufacturing, such as establishing datums, accounting for thermal expansion and tolerance-checking using 3D scans.

Activities:

Badminton: I enjoy badminton and train a minimum of thrice a week. I developed resilience throughout the year while training for getting into the university team next year, after missing out this year. I find it a great way to stay active with a good social aspect.

PC building: I have conducted a side-hustle in building, fixing and selling PCs. It helped me build problem-solving and commercial skills, especially when fixing PCs.

Hiking: I love hiking and exploring nature since it can both be a great place for introspection and for socialising with friends. Also, hard hikes such as the one I did in the Himalayas can be incredibly rewarding while being difficult and requiring resilience.